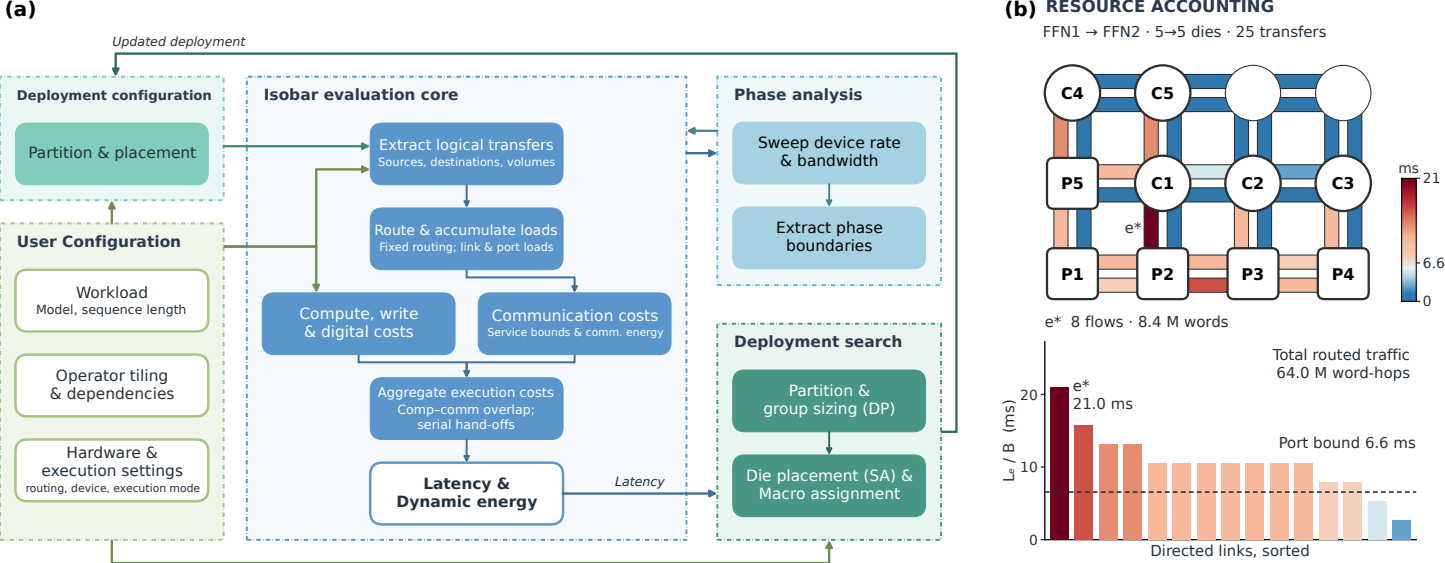
(a)
Updated deployment
Deployment configuration
Partition & placement
User Configuration
Workload
Model, sequence length
Operator tiling
& dependencies
Hardware &
execution settings
routing, device, execution mode
Isobar evaluation core
Extract logical transfers
Sources, destinations, volumes
Route & accumulate loads
Fixed routing; link & port loads
Compute, write
& digital costs
Communication costs
Service bounds & comm. energy
Aggregate execution costs
Comp–comm overlap;
serial hand-offs
Latency &
Dynamic energy
Latency
Phase analysis
Sweep device rate
& bandwidth
Extract phase
boundaries
Deployment search
Partition &
group sizing (DP)
Die placement (SA) &
Macro assignment
(b) RESOURCE ACCOUNTING
FFN1 → FFN2 · 5→5 dies · 25 transfers
C4
C5
P5
C1
C2
C3
e*
P1
P2
P3
P4
ms
21
6.6
0
e* 8 flows · 8.4 M words
Total routed traffic
64.0 M word-hops
e*
21.0 ms
Port bound 6.6 ms
L_e / B (ms)
20
10
0
Directed links, sorted